Anti-Corruption Base Intelligence

Item Configuration-Change Request

Expansion of Financial Data Access and International Anti-Corruption Integration V-1

Plantel: Centro Universitario de los Valles

Materia: Software Configuration Management

Nombre del profesor: *Omar Ali Zatarain Durán*

Grado: Tercer semestre

Prepared by: Hiram Gabriel Alonso Marquez

Index

Introduction.....2
Objectives.....2
Petitions:.....3
Item Configuration CR-002.....3
Item Configuration CR-003.....8
Item Configuration CR-004.....14

Introduction

Version V-1 CR-002: The client (Government) has streamlined the data acquisition strategy. Instead of generic manual API configuration for all data sources, a specific requirement has been introduced to allow authorized bureaucrats to retrieve financial movements directly by using their unique CURP (Clave Única de Registro de Población). This simplifies the onboarding process for specific financial audits but requires a new dedicated module for secure CURP validation and banking data retrieval.

Version V-1 CR-003: The government client has requested an expansion of the financial auditing capabilities. In addition to retrieving the employee's financial movements via CURP, the system must now obtain the financial details of their direct family circle.

Version V-1 CR-003: The client (Government) has established new diplomatic anti-corruption relations with Switzerland. To comply with international cooperation treaties, the system requires a new module dedicated to the secure exchange of information with the Swiss anti-corruption service. This module involves a proprietary middleware technology for which no current personnel in Mexico has expertise. Consequently, the project requires the hiring of a specialized team from Switzerland (supported by their government) and a significant extension of the project timeline

Objectives

1. CR-002: Establish the technical specifications for the Module 6: CURP Financial Gateway to validate government credentials (CURP).
2. CR-002: Expand Module 6: CURP Financial Gateway to query government registries for linked relatives based on the primary employee's CURP.
3. Establish the technical and legal framework for Module 7: Swiss Secure Bridge to enable data exchange with Swiss authorities.
4. CR-003: Hire specialized international personnel (Switzerland-based) to implement the proprietary middleware.
5. CR-003: Extend the project timeline by 6 months (26 weeks) to accommodate development, international testing, and compliance validation.
6. CR-003: Adjust the total project budget to $20,000,000 MXN to cover international consulting fees, government cooperation costs, and extended operational expenses.

Petitions:

CR-002.- The government (client) has identified that it only requires that a bureaucrat install the app using its CURP to obtain the details of financial movements. Therefore, the client requests a new module to obtain the bank details to start retrieving financial movements.

2.- The client requests to include the financial movements of direct relatives of government employees (couple, parents, brothers, sisters, children).

3.- The client has established new anti-corruption relations with Switzerland and requires a new module for exchanging information with the Swiss anti-corruption service. The new module requires implementing a different middleware, for which no programmer in Mexico is familiar.

Item Configuration CR-002

CR-002:

Change Request Description: Implementation of a bank linking and authentication module via CURP (Mexican National ID).

ID Change Request: 02

Request Date: April 20, 2026

Requester: Government

Impact on the Project:

Affected Modules:

1. New Module 6: CURP Financial Gateway (CURP Input, Validation, Bank Data Retrieval).
2. Module 1: Data Listener (Updates to consume financial feed from Module 6).
3. Module 5: Time Paradox (Updates to display specific financial movement alerts).

Planning and Timeline: Adds 2 additional weeks of work, distributed across Core Development (Backend connection to banking APIs) and Security & Anti-Forensics (Protection of CURP data). This extends the total timeline from 27 to 29 weeks.

Associated Risks: High risk. Requires isolation of the CURP module to prevent leakage if the device is compromised. Must ensure the app does not store the CURP in plain text.

Additional Effort Estimate: 60-70 hours of development (Backend Rust Dev for secure API consumption, Frontend for CURP input UI, and Senior Crypto for securing the PII pipeline).

Approval: Approved

A**dditional Effort Estimate**: An increase of 60-70 additional hours is estimated.

Updated Team Configuration

The change request is incorporated without adding new roles, see Table 1 Team:

#	Role	Level	Qty	Key Responsibilities (Updated)	Monthly Salary (MXN)	Total (6.75 meses)
1	**Sr. Crypto & Security**	Senior	1	E2EE, anti-forensics, **Secure handling of CURP/PII**, encryption of financial streams.	$68k – $75k$	**$482,625**
2	**Sr. Rust & Architecture**	Senior	1	Binary obfuscation, **Architecture for Module 6 (Banking API)**, secure data piping.	$65k – $72k$	**$462,375**
3	**Frontend Developer**	Mid	1	React, TS, **UI for CURP & dashboards**, visual graphs for bank movements.	$32k – $42k$	**$249,750**
4	**Backend Rust Dev**	Mid	1	Data Listener, **Parser for financial movements**, API consumption logic (Module 6).	$32k – $40k$	**$243,000**

5	**Integration / Mobile**	Mid/Jr+	1	Tauri 2.0, IPC, **Biometric integration for CURP access**, mobile builds.	$28k – $38k$	**$222,750**
6	**Full-Stack / Support**	Jr/Mid	1	General FE/BE support, testing of the **CURP login flow**.	$25k – $35k$	**$202,500**
7	**Tester / Security Eng.**	Senior	1	Pentesting, **Validation of PII protection**, preventing CURP leaks in logs/memory.	$45k – $55k$	**$337,500**
8	**PM / Scrum Master**	—	1	Coordination, backlog for **Module 6**, timeline extension oversight.	$22k – $28k$	**$168,750**

Table 1. Team

Personnel Cost Summary (Estimated Average):

2 Senior Developers: ≈ MXN 947,250

4 Mid/Junior Developers: ≈ MXN 921,875

1 Senior Security Tester: ≈ MXN 337,500

1 Part-Time Project Manager: ≈ MXN 170,100

Total Personnel Cost for 6.75 Months (27 weeks) ≈ MXN 2,376,725 – 2,650,000 (Increase of ~MXN 175,000 – 200,000 due to the 2-week extension and increased complexity).

Updated Material Resources

Status: Approved, 4/22/2026

Changes: Addition of secure hardware token simulators for testing banking APIs.

Updated Project Phases

Adds 2 weeks total, distributed across Initiation (requirements for banking APIs), Core Development (building Module 6), and Security (PII protection). Timeline extended to 27 weeks, see Table 2 Project Phases:

Phase	**Weeks**	**% of Total Time**	**Main Activities (Updated)**	**Roles with Highest Effort**	**Important Notes**
1. Initiation & Definition	1–2	~7%	Kick-off, **Requirements gathering for Banking/CURP APIs**, security design for PII.	PM, Sr. Rust, Sr. Crypto	**New Module Requirements.**
2. Crypto Research & Prototyping	3–5	~11%	Primitive validation, **Prototyping secure CURP storage/in-memory handling**.	Sr. Crypto, Sr. Rust	Preventing PII leaks.

3. Core Development (Rust)	4–15	~44%	Rust core implementation: entities, Data Listener; **Dev of Module 6: CURP Gateway & Financial Parser**.	Sr. Rust, Mid Rust	**+2 weeks added for Module 6.**
4. Frontend Development	6–16	~37% (parallel)	Dashboard, timelines, **CURP Input Screen, Financial Movement Charts**.	Mid Frontend, Jr. Full-Stack	UI for Bank details.
5. Multiplatform Integration	12–18	~30% (parallel)	Rust ↔ React bridge, **Biometric Auth integration for CURP module**, mobile builds.	Mid Integration, Mid Frontend	Secure access flow.
6. Security & Anti-Forensics	10–21	~58% (cross-cutting)	E2EE implementation, **Sanitization of logs for CURP data**, sandbox detection.	Sr. Crypto, Sr. Tester	**Critical for PII compliance.**

7. Intensive Testing + QA	16–24	~44%	Unit, integration, **Penetration testing on CURP module**, validation of financial data accuracy.	Sr. Tester, all developers	**Testing banking integration.**
8. Polishing, Performance & Beta	21–25	~22%	Optimizations, **Real device testing with banking simulators**, closed beta.	Entire team	Final polish of Module 6.
9. Closure & Delivery	25–27	~11%	Final documentation, **Delivery of CURP module specs**, training, final packaging.	PM, Sr. Rust, Tester	Handover of financial features.

General Guidelines Updated:

1. The current project stage (phases 1-9 of the updated 27-week timeline).
2. The scale of the change (CURP integration requires High security priority).

Item Configuration CR-003

ID Item Configuration: 03

Change Request Description: Expansion of Financial Scope to Direct Relatives (Family Group)
ID Change Request: 003

Request Date: April 15, 2026
Requester: Government

Affected Modules:

- Module 6: CURP Financial Gateway (New logic for "Family Tree" retrieval and multi-profile fetching).
- Module 1: Data Listener (Increased capacity to handle multiple simultaneous data streams per case).
- Module 2: Network Analysis (Updates to visually distinguish family nodes and financial flow between relatives).
- Module 5: Time Paradox (Algorithms updated to detect "shared anomalies" across family accounts).

Planning and Timeline:

Adds 1 additional week of work, focused on Backend logic for relationship parsing and Frontend updates for family tree visualization. This extends the total timeline from 27 weeks to approximately 30 to 31 weeks. Considered a High priority.

Associated Risks: Increased API rate consumption (fetching data for 3-5 additional people per case).

Approval: Approved

Additional Effort Estimate: 40-50 hours of development (Backend Rust Dev for multi-query logic, Frontend for relationship visualization, and Data Science logic for cross-correlation).

Updated Team Configuration: The change request is incorporated without adding new roles. The team remains at 8 people, see Table 1. Team:

#	Role	Level	Qty	Key Responsibilities (Updated)	Monthly Salary (MXN)	7.5-Month Total (MXN)

1	**Sr. Crypto & Security**	Senior	1	E2EE, **Secure storage of multiple PII profiles (Family Group)**, preventing data leakage between linked profiles.	68k – 75k	$536,250
2	**Sr. Rust & Architecture**	Senior	1	Binary obfuscation, **Architecture for recursive data fetching (Employee -> Relatives)**, optimizing DB queries for families.	65k – 72k	$513,750
3	**Frontend Developer**	Mid	1	React, TS, **Family Tree Visualization Component**, UI for toggling between employee and relative financial views.	32k – 42k	$277,500
4	**Backend Rust Dev**	Mid	1	Data Listener, **Implementation of relationship mapping logic (Parser for Civil Registry)**, multi-threaded data fetching.	32k – 40k	$270,000
5	**Integration / Mobile**	Mid/Jr +	1	Tauri 2.0, IPC, **Performance optimization for rendering larger**	28k – 38k	$247,500

				datasets (families) on mobile devices.		
6	**Full-Stack / Support**	Jr/Mid	1	General FE/BE support, **Testing of edge cases (e.g., orphaned records, missing family links)**.	25k – 35k	$225,000
7	**Tester / Security Eng.**	Senior	1	Pentesting, **Validation that one relative's compromise does not expose the others' data**, testing cross-alerting logic.	45k – 55k	$375,000
8	**PM / Scrum Master**	—	1	Coordination, backlog management for **Family Module features**, timeline extension oversight.	22k – 28k	$187,500

Table 1. Team

Personnel Cost Summary (Estimated Average):

2 Senior Developers: ≈ MXN 1,050,000
4 Mid/Junior Developers: ≈ MXN 1,020,000
1 Senior Security Tester: ≈ MXN 375,000
1 Part-Time Project Manager: ≈ MXN 187,500
Total Personnel Cost for 7.5 Months (30-31 weeks) ≈ MXN 2,632,500 – 2,950,000 (Increase adjusted for the extended timeline to 30-31 weeks).

Updated Material Resources
Status: Approved, 5/1/2026

Changes: Procurement of anonymized "Family Group" datasets for testing relationship parsing algorithms.

Updated Project Phases
Timeline adjusted to span 30-31 weeks to accommodate the integration and testing of the Family Group logic, see Table 2. Project Phases:

Phase	Weeks	% of Total Time	Main Activities (Updated)	Roles with Highest Effort	Important Notes
1. Initiation & Definition	1–2	~6%	Kick-off, requirements for Banking & Family Mapping APIs, security design.	PM, Sr. Rust, Sr. Crypto	Initial planning.
2. Crypto & Data Research	3–5	~10%	Prototype secure storage for Multiple PII Profiles, cross-profile isolation logic.	Sr. Crypto, Sr. Rust	Security of linked data.
3. Core Development (Rust)	4–18	~48%	Rust core: Data Listener; Dev of Family Mapping & Recursive Financial Parsers.	Sr. Rust, Mid Rust	**+3 weeks for Family logic.**

4. Frontend Development	6–21	~48%	Family Tree Visualization, toggles for linked accounts, financial charts.	Mid Frontend, Jr. Full-Stack	Complex UI components.
5. Multiplatform Integration	13–23	~32%	Bridge optimization, Biometric Auth for family-shared devices, mobile builds.	Mid Integration, Mid Frontend	Performance on mobile.
6. Security & Anti-Forensics	10–25	~48%	E2EE, Sanitization of family-linked logs, Preventing PII leaks between profiles.	Sr. Crypto, Sr. Tester	**Critical: Isolation.**
7. Intensive Testing + QA	18–28	~32%	QA for Family Edge Cases (broken links), Pen-testing on linked data gateway.	Sr. Tester, all developers	Full integration test.
8. Polishing & Beta	24–29	~16%	Performance tuning for large data trees, Closed Beta with family-group scenarios.	Entire team	UX/UI refinement.

9. Closure & Delivery	30–31	~6%	Final documentation, Handover of Family Module specs, project sign-off.	PM, Sr. Rust, Tester	Final packaging.

Table 2. Project Phases

General Guidelines Updated:

The current project stage (phases 1-9 of the updated 30-31 week timeline).
The scale of the change (Family integration requires High logic priority).

Item Configuration CR-004

ID Item Configuration: 04

Change Request Description: Integration of Swiss Anti-Corruption Service & Specialized Middleware

ID Change Request: 004

Request Date: May 20, 2026

Requester: Government (International Relations Dept.)

Impact on the Project:

Affected Modules:

New Module 7: Swiss Secure Bridge (Middleware integration, Protocol adaptation).
Module 1: Data Listener (Export/Stream compatibility with Swiss formats).
Module 5: Time Paradox (International alerts and cross-border anomalies).

Planning and Timeline:
Adds a 6-month extension (26 weeks) to the previous timeline of 31 weeks. The total project duration is now approximately 57 weeks (~14 months). This extension is dedicated exclusively to the international integration phase.

Associated Risks: Coordination between the Mexican team, Swiss team, and both governments.

Approval: PENDING APPROVAL (Awaiting authorization of the $20,000,000 MXN budget).

Additional Effort Estimate: 800+ hours of specialized international consulting (Swiss Middleware Architects) + 1,000+ hours of Mexican team support for integration.

Updated Team Configuration
The change request incorporates 3 new specialized roles hired from Switzerland (or contracted via the Swiss government liaison). The core Mexican team remains (8 people) but is extended for an additional 6 months, see Table 1. Team

#	Role Level	Origin	Qty	Key Responsibilities (Updated)	Monthly Cost (MXN)	Total Cost (14 Months)
MEXICAN TEAM						
1	Sr. Crypto & Security	MX	1	E2EE adaptation for Swiss protocols; FADP compliance.	$70,000	$980,000
2	Sr. Rust & Architecture	MX	1	Refactoring architecture for **Middleware Plugin** injection.	$68,000	$952,000

3	Frontend Developer	MX	1	UI for International Requests; Swiss status indicators.	$38,000	$532,000
4	Backend Rust Dev	MX	1	Data serialization for Swiss formats; API adapters.	$38,000	$532,000
5	Integration / Mobile	MX	1	Testing bridge connectivity on local devices.	$35,000	$490,000
6	Full-Stack / Support	MX	1	Support for international documentation; bug fixing.	$32,000	$448,000
7	Tester / Security Eng.	MX	1	**Compliance Testing:** Validation of Swiss-Mexican data transfer.	$55,000	$770,000
8	PM / Scrum Master	MX	1	Coordination of **Bi-national teams** and gov reporting.	$28,000	$392,000

SWISS TEAM						
9	**Swiss Middleware Arch.**	**CH**	**1**	**Implementation of "SwissGateway" middleware;** onsite training.	**$350,000**	**$4,900,000**
10	**Int. Liaison / Lead**	**CH**	**1**	**Bridge between MX Devs and Swiss Gov IT; Requirements.**	**$180,000**	**$2,520,000**
11	**Compliance Officer**	**CH**	**1**	**Legal oversight of data flows; GDPR/FADP certification.**	**$200,000**	**$2,800,000**
TOTAL			**11**	**Combined Global Workforce**	**$1,074,000**	**$15,316,000**

Table 1. Team

Personnel Cost Summary (Estimated Average):

- Mexican Team (14 months): ≈ MXN 5,096,000
- Swiss Consulting Team (14 months): ≈ MXN 10,220,000
- Government/Legal Fees & Licensing (Est.): ≈ MXN 4,684,000
- Total Project Cost: ≈ MXN 20,000,000 (As requested for approval).

Updated Material Resources Status: Pending Approval Changes:

- Procurement of proprietary "SwissGateway" SDK licenses.
- Secure dedicated servers for the hand-off point (Mexico-Switzerland).

- Travel budget for Mexican team leads to Zurich for knowledge transfer (Weeks 1-2 of extension).

Updated Project Phases:

The project adds a final "Phase 10: Swiss Integration & Compliance" lasting 6 months (26 weeks), see Table 2. Project Phases.

Phase	Weeks	Timeline	Main Activities (Updated)	Roles with Highest Effort	Important Notes
1–9. Core & Family Module	1–31	Months 1–7.5	Local features, Family Tree, secure PII storage, and biometric auth.	Mexican Team	**Base System Completion.**
10. Swiss Protocol Initiation	32–35	Month 8–9	**Kick-off with Swiss Gov;** Legal framework definition; **Middleware Environment Setup.**	PM, Swiss Liaison, Swiss Arch.	Diplomatic clearance required.
11. Middleware Implementation	36–43	Month 10–11	**Coding of SwissGateway Adapter;** Data mapping (MX API -> CH Schema); **Handshakes.**	Swiss Arch., Sr. Rust, Sr. Crypto	Dependency on Swiss Expert.

12. Int. Integration & QA	44–49	Month 12–13	**Cross-border data testing;** Compliance audits (GDPR/FADP); **Bridge load testing.**	Compliance Officer, Sr. Tester	**Critical Validation Phase.**
13. Final Polish & Delivery	50–53	Month 14	**Training for International Module;** Final sign-off from both Governments; **Handover.**	PM, Swiss Liaison, All Leads	**Project Closure.**

Table 2. Project Phases

General Guidelines Updated:

1. The current project stage extends significantly. The core local functionality remains the priority, but the final 6 months are blocked exclusively for Swiss integration.
2. The scale of the change is critical, see Table 3. Project Scale, due to the international treaty obligation.

Priority Level	Description
CRITICAL	**Mandatory immediate implementation.**
HIGH	**High priority.**

MEDIUM	**Desirable improvement.**
LOW	**Optional / Secondary.**

Table 3. Project Scale